

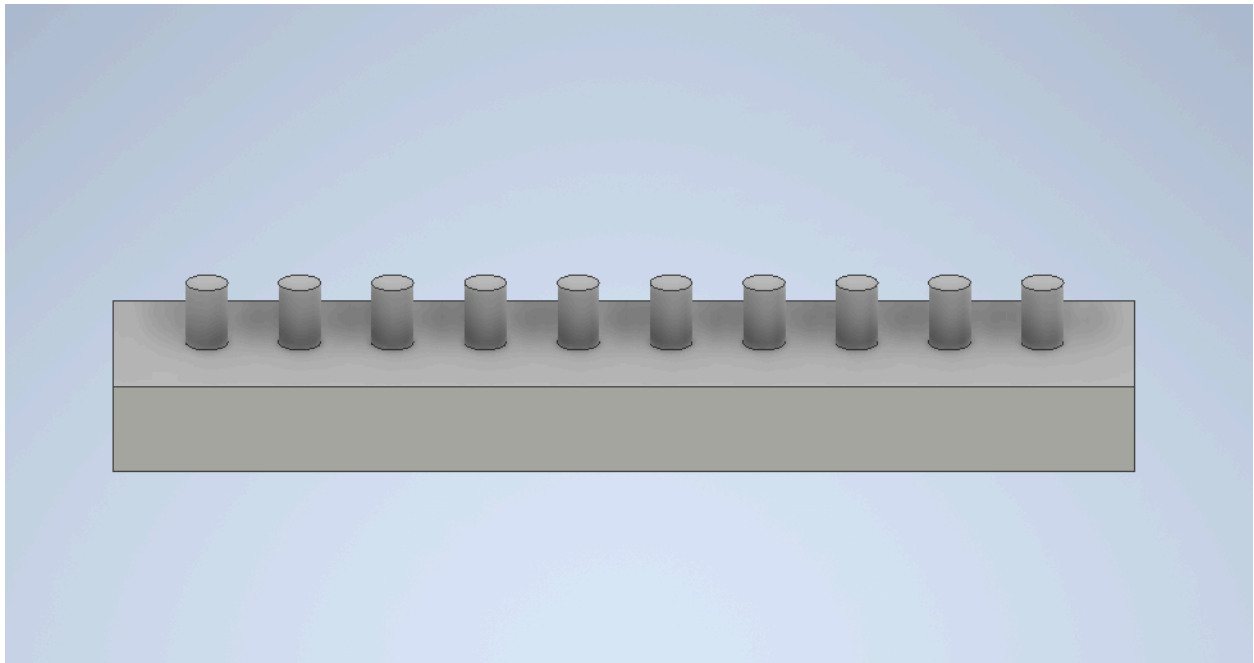
MyDesign Portfolio

Element G: Construction of a Testable Prototype

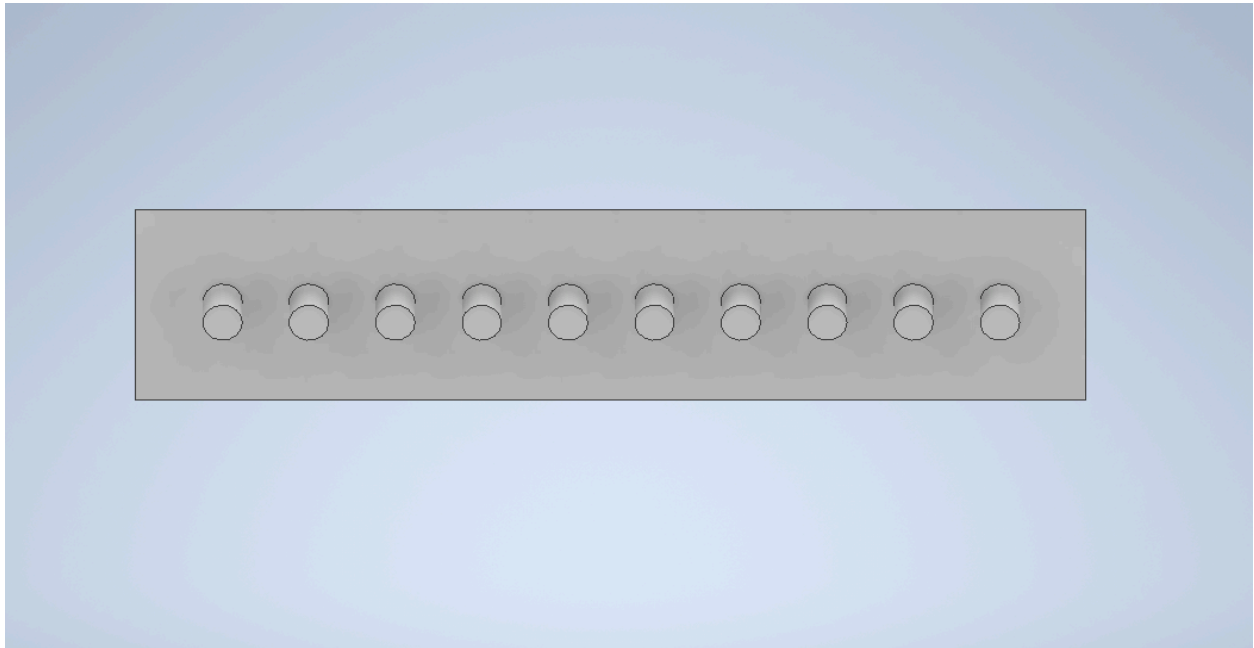
Final Prototype Iteration Explanation (G1)

Prototype Construction of Design #1:

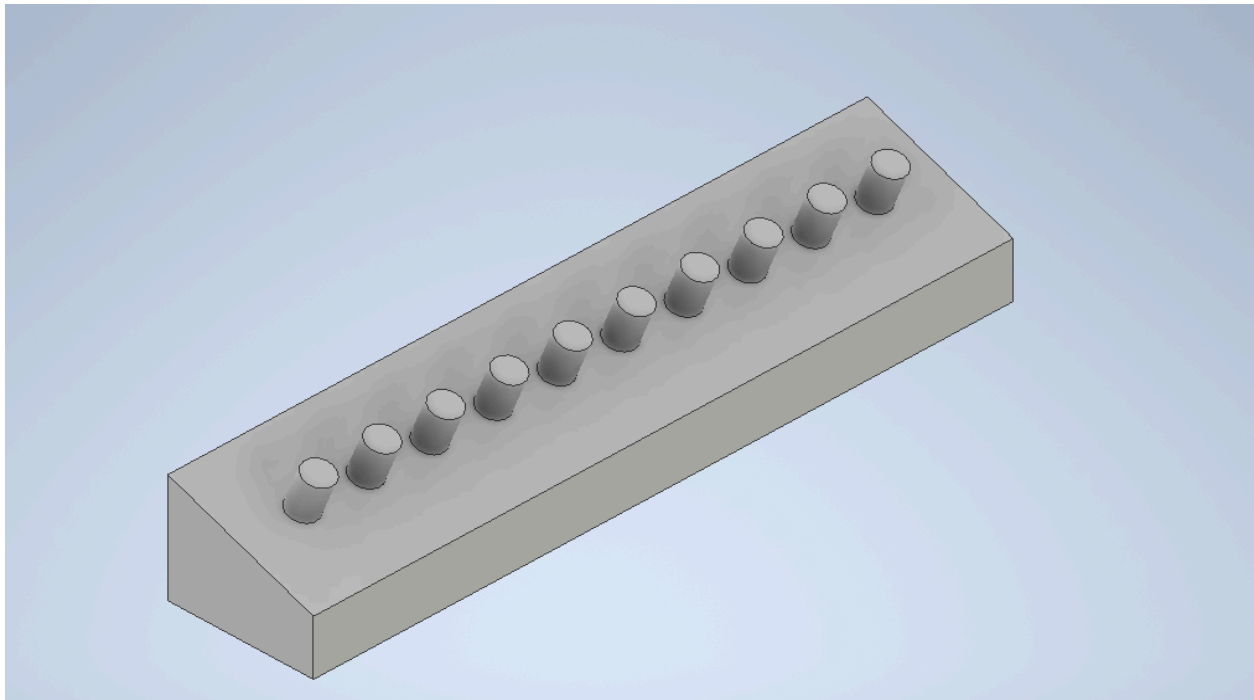
Front:



Top:



Perspective:

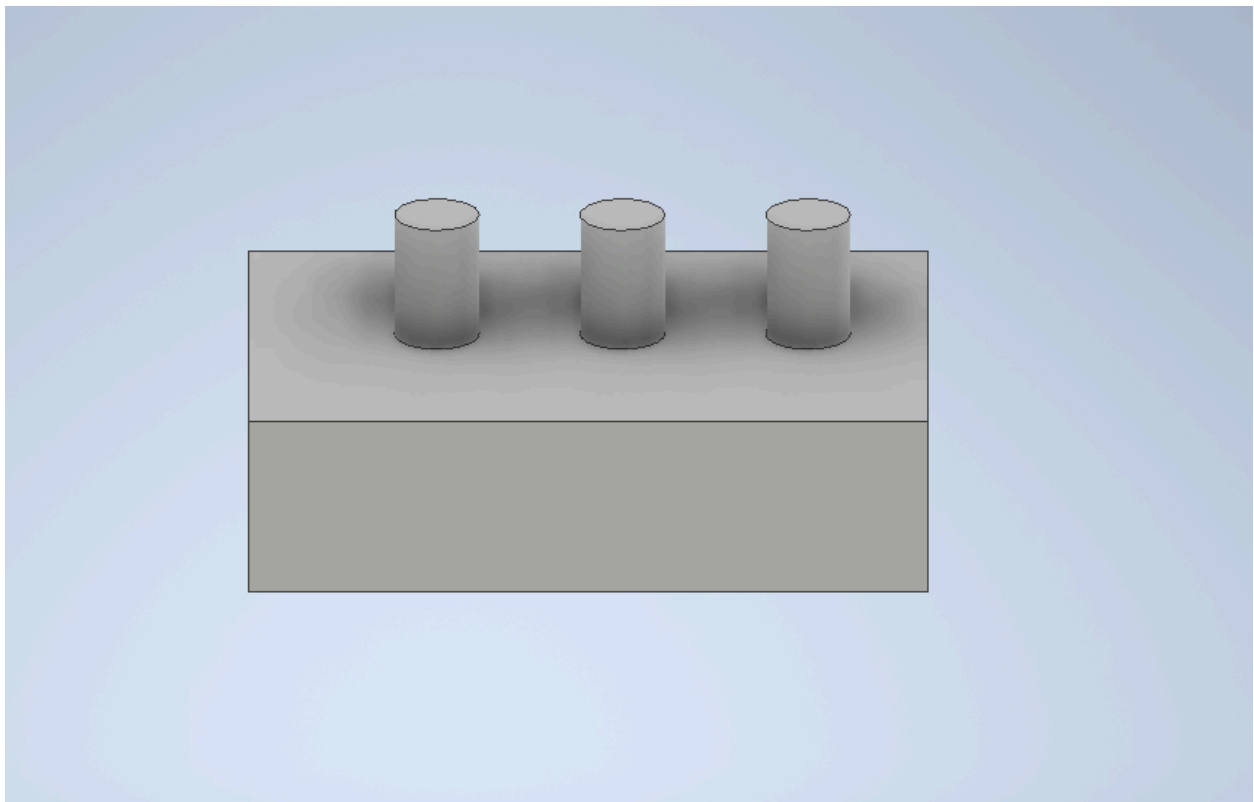


Date: 4/4/2025

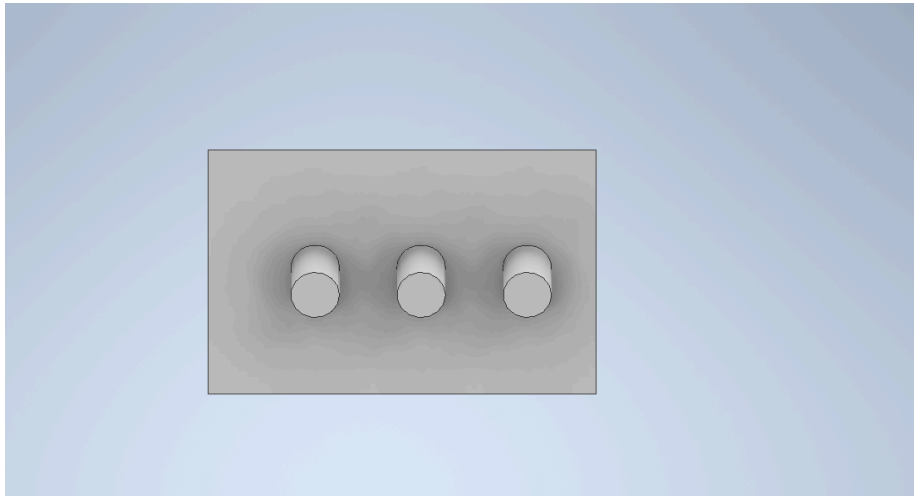
The CAD version of the prototype incorporates the design ideas that are required, but this serves to test the ability that the project will have to hold up with the 5 pegs which are used on the board, the Socket holders on the top are not sized properly to hold sockets and only serve to make the weight test on the pegboard to be more accurate

Prototype Construction of Design #2

Front:



Top:



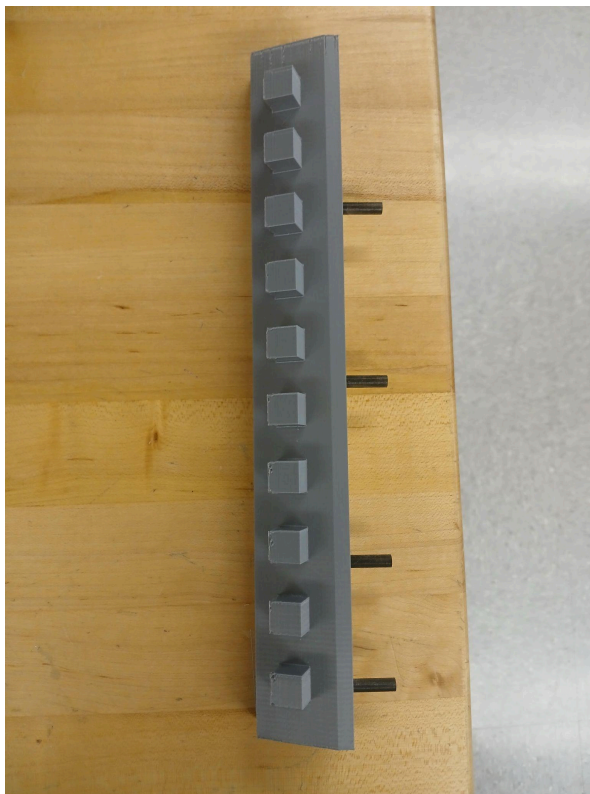
Physical prototype of #2



Date: 4/4/2025

A smaller version of prototype 1 which is used to test $\frac{1}{3}$ of the capabilities that the final product should have. This also allows for less time for 3D print completion and material cost for the prototype

Prototype 3:



Date: 5/10/2025

Description: A fully plastic 3D printed model with the pegs also being plastic. Was able to pass almost all of the tests.

Prototype 4:



Date: 5/19/2025

Description: Final prototype with metal pegs that had used epoxy to secure them. The sockets also fit on the holders that are on. Passed all tests.

	Details	Tools and Manufacturing Methods used
Step 1	Our first CAD prototype was based almost exactly on the blueprint, but has 2 more peg slots	3D printer and CAD software (Inventor)
Step 2	Prototype sized down to $\frac{1}{3}$ of the original size for optimized testing	3D printer and CAD software (Inventor)
Step 3	Full Size Prototype with 3D printed pegs to test its strength on the pegboard at full size	3D printer and CAD software (Inventor)
Step 4	Full Size final prototype with changes to fit on the pegboard better and metal pegs to complete testing procedures	3D printer, Metal Lathe and CAD software (Inventor)
Step 5	Final Design x3	3D printer, Metal Lathe and CAD software (Inventor)
Step 6		

Step 7		
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Material name	Description	Where material was purchased	Cost per unit	# units	Total cost per item
3D filament	Plastic string that is melted in a 3D printer and placed to form objects.	Was not	0	0.15 lbs	0
Metal rod	A 36x1/4 in cylindrical metal rod	Home depot	\$4.98	1	\$4.98

Detailed sketches, which might include CAD if you have learned it in your class:

Construction of the Final Prototype (G2)

Design Requirement	How will this prototype allow you to collect objective, measurable data?
Pegboard compatible	Has to be able to fit on a pegboard
Holds all sockets	Has the correct amount of slots for all sockets

Right size for sockets	Each socket has a slot that fits the opening
Sturdy to hold weight	The holder can hold the weight of all the sockets and not fall
transportable	Able to go on and off the pegboard

Testing or Modeling of Attributes (*G3*, *G4*) and Addressing Non-Testable Attributes (*G4*)

Attribute	Description	Describe your prototype's functionality to test this attribute
Metal pegs	Straight metal pegs to attach to pegboard	Strength and ability to be transported
slots	10 square slots	Square slots to match the sockets and has enough to accompany all the sockets

Addressing Non-Testable Attributes (G4)

We are able to test all our attributes.

